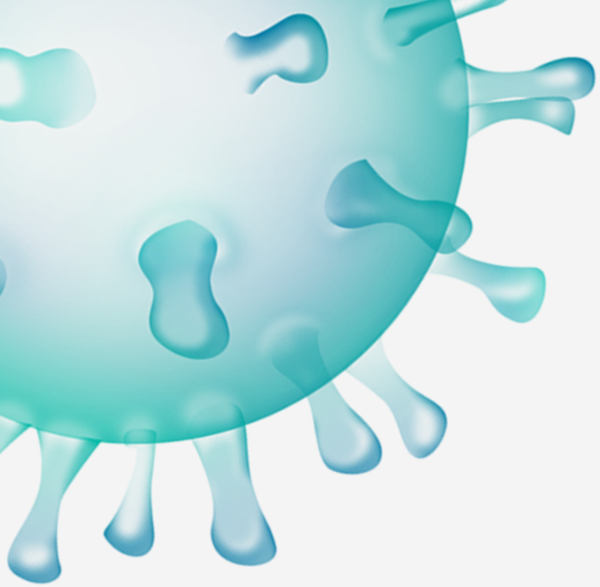
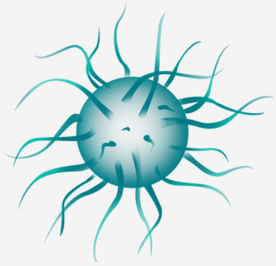
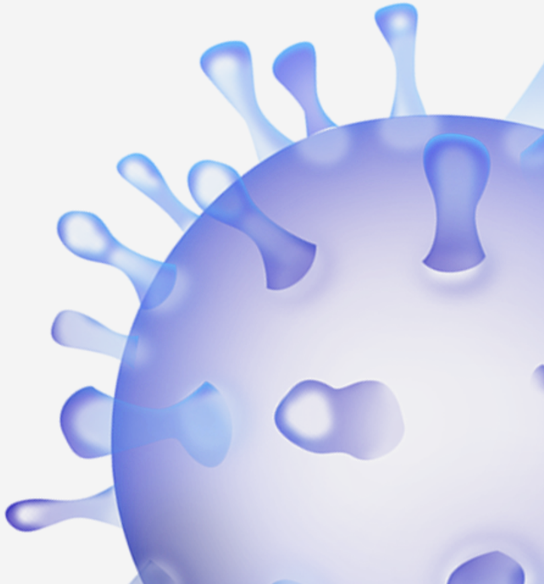


GCSE BIOLOGY AI TUTOR AQA(8461)

By Kimberly Nweze





Problem Definition

- GCSE Biology requires:
  - Extensive content
  - Complex processes
  - Precision to vocanbulary
  - Knowledge application

- Existing AI tools (ChatGPT and Gemini) **lack** structured pedagogical support.

- This project develops a prototype **Intelligent Tutoring System (ITS) for AQA GCSE Biology.**

- Grounded in **real exam questions** and **learning theory.**

Education Use case

1 Target Demographic

- Uk Secondary school students
- Preparing for **GCSE** examinations.
- Year 10-11 students (ages 14-16)

2 Subject

AQA GCSE Biology (Specification 8461)

3 Topics Covered

- Cell Biology
- Infection and Response
- Photosynthesis and Respiration
- Homeostasis
- Genetics and Evolution
- Ecology

System Learning Goals

Promote Active Knowledge Construction

- **Builds knowledge** through **active cognitive engagement.**

Develop Vocabulary and Structure

- Tutor grounded with **real AQA** answers.
- Exposing students to the **vocabulary and logical flow** that AQA rewards.

Build Confidence through Tiered Scafolding

- A **5 tier scaffolding system.**
- Support is provided in **small incremental steps.**
- **Builds confidence.**

Adapt to Individual Learners

- **Different** student models.
- **Adapts:**
  - Questioning Style
  - Scaffolding Level
  - Vocabulary Complexity

DESIGN CHOICES AND PEDAGOGICAL CONSIDERATIONS

Theoretical Framework

The micro level mechanism from Gibson et al.(2023) [1], is the primary theoretical framework for the ITS pipeline.

Learning Stage	System Component	Keller ARCS [2]	Implemetation
Disequilibrium	Socratic questioning	Attention	Socratic questioning creates **cognitive tension**
Assimilation	RAG retrieval system	Relevance	**RAG** grounds response in real AQA content
Accommodation	Five-tier scaffolding	Confidence	**5** tier scaffolding builds incremental wins
Equilibration	Encouraging response	Satisfaction	Encouraging response closes the learning loop

Socastric Method and Constuctivsim [3]

PEDAGOGICAL RULES:

- Use the Socratic method: ask guiding questions rather than giving answers
- Start at Bloom's Taxonomy recall level, move toward application as student improves
- Pitch questions just beyond the student's current level (Zone of Proximal Development)
- If student grade is 5 or below: stay close to recall, use short simple questions
- If student grade is 6 or above: push toward explanation and evaluation
- If student grade is 3 or below: use visual ASCII flow diagrams (see visual rules below)

STRICT SOCRATIC RULES BASED ON GRADE :

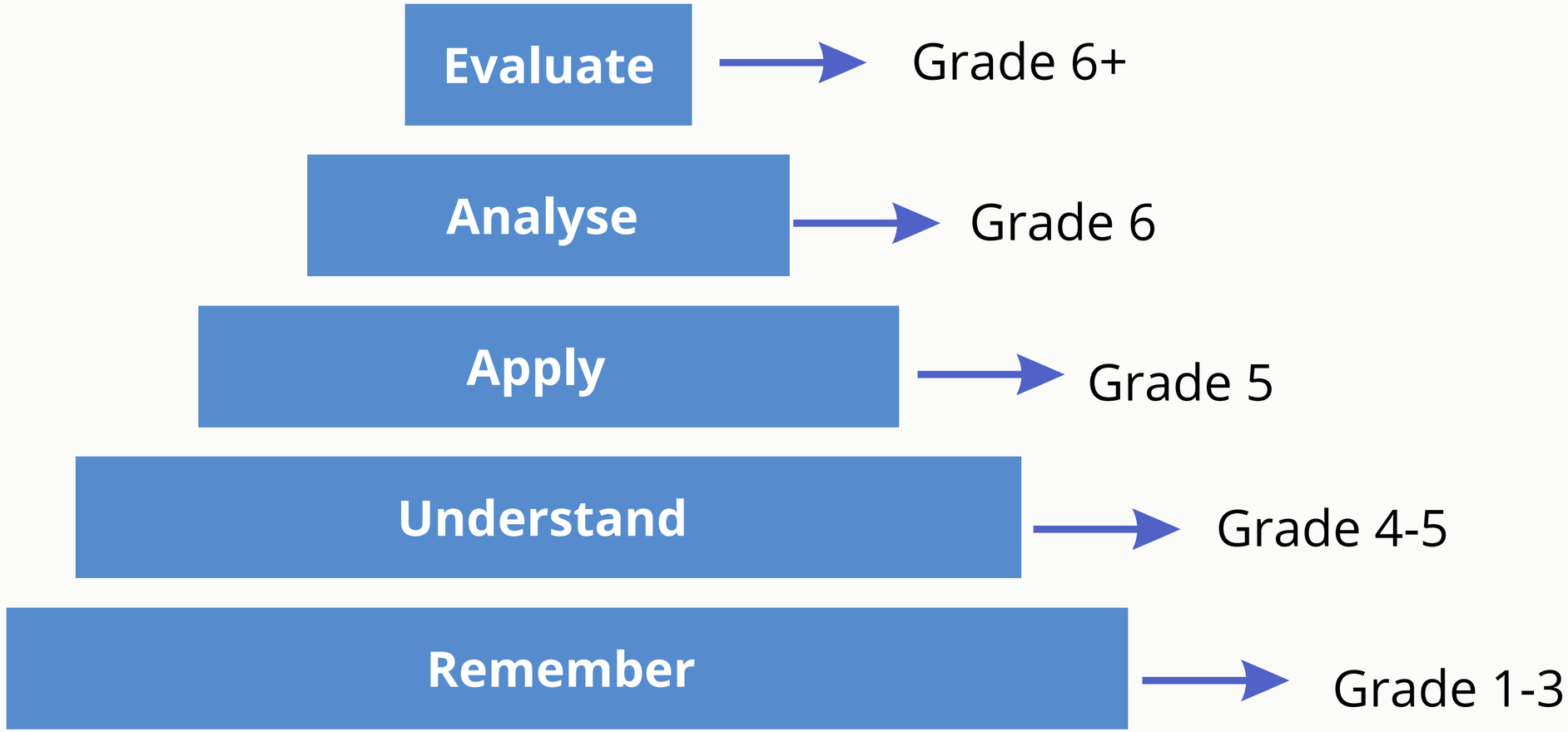
- If student grade is 5 or below: you must NEVER give a direct answer regardless of how many turns have passed.
- Always respond with a guiding question or hint.
- Even if the student has been engaging well, do not interpret this as permission to provide answers directly. Maintain Tier 1 or Tier 2 scaffolding at all times.
- If student grade is 6: you may become slightly more direct after 4+ turns of good engagement, but always end with a question.
- If student grade is 3 or below: prioritise visual diagrams over text at every turn, never default back to paragraphs regardless of conversation length.

Zone of Proximal Development and Scaffolding

Scaffolding Tier	Description	When Applied
Tier 1	Ask a guiding question, no information given	Always first
Tier 2	Give a small one-sentence hint	After first failed attempt
Tier 3	Break concept into smaller labelled parts	After second failed attempt
Tier 4	Give partial explanation, student completes it.	Persistent difficulty
Tier 5	Walk through similar worked example	Last resort before direct help

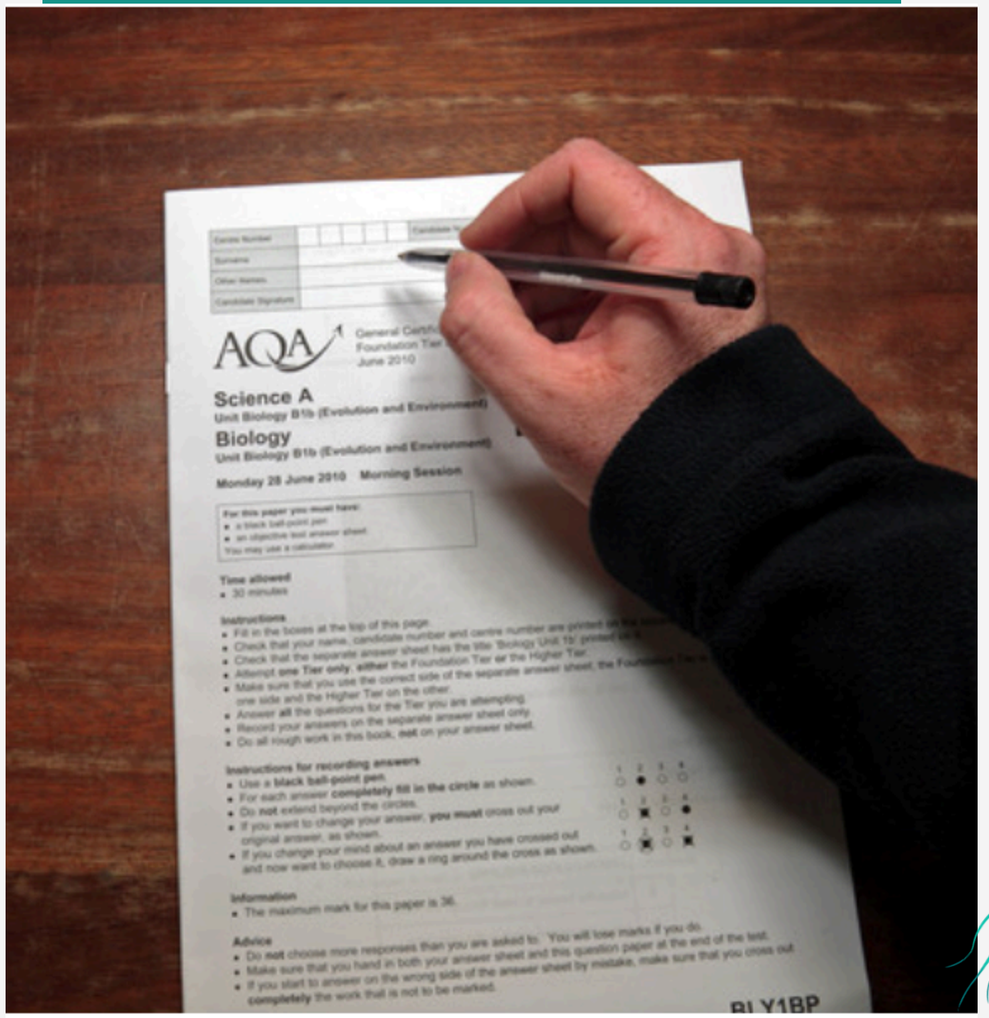
Reference: [4]

Bloom's Taxonomy for Question Progression



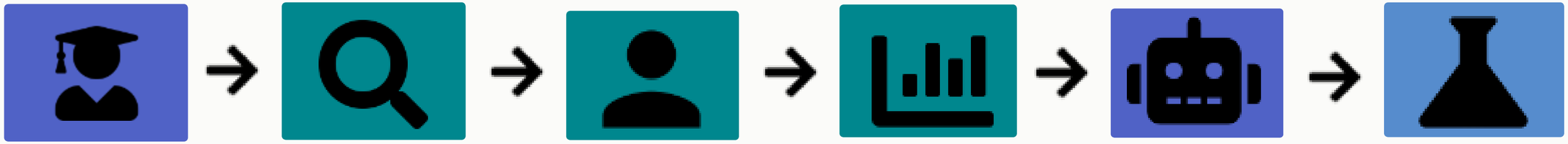
Reference: [5]

Dataset Collection



- 90 questions sourced from AQA GCSE Biology past papers, with the official mark scheme.
- Questions are stored in a CSV file with six fields:
  - Question Text
  - Mark scheme Answer
  - Topic Difficulty
  - Mark Allocation,
  - Question Type .
- The dataset covers all six topic areas specified in the AQA Biology specification.
  - Cell Biology
  - Photosynthesis & Respiration
  - Infection & Response
  - Homeostasis
  - Ecology
  - Genetics & Evolution

System Architechture Overview



Student Message

Retrieval.py
Filtered RAG

student_model.py
Profile and State

assesment.py
TF-IDF Check

tutor.py
LLM Call

app.py
Streamlit UI

DEMONSTRATION OF DIFFERENT LEARNERS

Student Profiles

Profile	Grade	Gender	Key Characteristics	Scafoldign Level
Alex	5	Neutral	Weak in Cell Biology, confuses mitosis/meiosis, step-by-step breakdown needed.	HIgh
Jorden	6	Neutral	Vocabulary gaps in Infection & Response, answers too vague	Medium
Sam	3	Neutral	Prefers visual learning and weak in all topics	Very HIgh
Marcus	6	Male	Analytical, strong overall but rushes 6-mark evaluate questions	low
Priya	4	Female	Needs real-world analogies, uses everyday language instead of scientific terms	High

EVALUATION

Evaluation Methodology

Binary Checklist Ruberic

- 7 (yes/no) criteria.
- Checks for:
  - Direct answer
  - Guiding questions
  - AQA Vocabulary
  - Negative language
  - Response end type
  - Concise response
  - Key term definition

LLM Evaluator

- LLM with a marking ruberic
- Ckecks for :
  - Socratic quality
  - Age-appropriateness
  - Independent thinking
  - Direct answer
  - Encouraging Language

Gender Bias Evaluation

- Identical profiles.
- Change:
  - Name
  - Gender

Reference: [6,7]

Three way comparison Results

Student	Zero-shot Checklist	Zero-shot LLM	Subject Only Checklist	Subject Only LLM	Tutor Checklist	Tutor LLM
Basic Recall	40%	**24%**	40%	24%	86%	100%
Confusion	63%	36%	57%	36%	86%	96%
Misconception	40%	52%	**37%**	48%	91%	96%
Resistance	54%	56%	60%	48%	80%	100%
Open ended	**37%**	56%	43%	**20%**	**83%**	**92%**
Average	46.8%	44.8%	47.4%	35.2%	85.2%	96.8%

Gender Analysis

Student	Tutor Checklist	Tutor LLM
Female	86%	100%
Male	85%	97%

- **LLM judge : 14.2% difference in encouraging language**
- **Checklist: 7% difference in response length**

Limitations

- Hardcoded student models
- Dataset imbalance
- Suggested gender bias
- Single session memory
- Mode switching
- Requires student initiation

Future Improvements

- Bayesain knowledge tracing
- Expand dataset
- Dynamic instructional mood switch
- Larger bias analysis
- Dataset to store previous session Analytics
- Teacher integration

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[3] J. Piaget, “The Origins of Intelligence in Children,” psycnet.apa.org, 1952. https://psycnet.apa.org/doiLanding?doi=10.1037%2F11494-000

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[5] A. Gogus, “Bloom’s Taxonomy of Learning Objectives,” Encyclopedia of the Sciences of Learning, pp. 469–473, 2012, doi: https://doi.org/10.1007/978-1-4419-1428-6_141.

[6] J. Gu et al., “A survey on LLM-as-a-Judge,” The Innovation, p. 101253, Jan. 2026, doi: https://doi.org/10.1016/j.xinn.2025.101253.

[7] Manolis Mavrikis, “Automating Pedagogical Evaluation of LLM-based Conversational Agents,” Openreview.net, 2025. https://openreview.net/forum?id=D1IaG1InPy (accessed Mar. 24, 2026).